Disha Shetiya

AI Engineer

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Summary

AI Engineer (M.S. Computer Science) with 2+ years building and evaluating LLM and RAG systems end to end, from retrieval design and prompt grounding through self-hosted inference and deployment. Built a document-intelligence RAG pipeline (Llama 3, FAISS) across 25+ RFPs, migrated inference to NVIDIA DGX Spark, and used quantitative evaluation (precision/recall/F1, hit@k, gold-label sets) to drive model, retrieval, and prompt decisions. Comfortable making accuracy, latency, and cost tradeoffs on ambiguous problems alongside engineers and stakeholders.

Skills

LLM & RAG: Llama 3, Qwen2.5, LangChain, FAISS, ChromaDB, embeddings and similarity search, prompt engineering, grounding/guardrails, LLM-as-judge, retrieval metrics (hit@k, MRR)
ML & AI: PyTorch, scikit-learn, Hugging Face, model evaluation (precision/recall/F1, confusion matrix)
Languages: Python, SQL, TypeScript, JavaScript
Data: DuckDB, PostgreSQL, data preprocessing & quality checks, ETL pipeline design and monitoring, Airflow
MLOps & Infrastructure: Docker, self-hosted GPU inference (DGX Spark), TorchServe, Jenkins CI/CD, Git
Full Stack: Next.js, React, React Native, Node.js, FastAPI, REST APIs, Prisma, Angular
Computer Vision: RetinaNet, ResNet-50, FPN, OpenCV, ORB-SLAM, GTSAM, Open3D, NVIDIA Jetson Nano

Work Experience

AI Developer, Angstrom LLC, Chicago — Jul 2025 - present

Brilliant Builders: Presented at Chicago Build Expo [Demo]

- Architected and delivered an end-to-end document-intelligence RAG pipeline (self-hosted Llama 3, FAISS, BAAI/bge-small-en), using Claude Code for agent-driven scaffolding and prompt iteration while owning architecture and review; migrated inference to NVIDIA DGX Spark, cutting review time ∼85% across 25+ RFPs and supporting 15+ contract wins.
- Building a research pipeline over a 1.3M-record procurement corpus (DuckDB) using self-hosted Qwen2.5-32B on DGX Spark to extract structured fields with evidence quotes; designed a gold-label evaluation (Wilson confidence intervals, inter-annotator Cohen's κ) to correct model-derived counts for classifier error.
- Led model selection for a 54-class construction safety detection system (RetinaNet + ResNet-50 + FPN), choosing RetinaNet over YOLO for class-imbalance and multi-scale tradeoffs; achieved 0.45 mAP@0.5 and deployed via TorchServe for automated compliance reporting.
- Built a real-time SLAM pipeline (ORB-SLAM, GTSAM) on NVIDIA Jetson Nano under strict compute constraints, achieving 15 FPS live pointcloud reconstruction accurate to ±2m for a wearable safety device.
- Identified a field data-collection bottleneck across 8 construction units (709 scope items) running on paper-based tracking, and owned the program end-to-end, architecting and shipping a full-stack replacement (Next.js, Node.js, React Native, Docker, PostgreSQL), cutting RFI turnaround from 3-4 days to under an hour.

Data Analyst, Innovise Technology Inc, Remote — Jan 2025 - Jul 2025

- Rebuilt legacy batch-processing healthcare ETL pipelines in Python + SQL with automated validation and instrumented monitoring, reducing processing time ∼25%, eliminating ∼60% of manual errors, and improving executive reporting speed ∼30%.

Machine Learning Engineer, University of Illinois, Springfield Feb 2024 – Jan 2025

- Owned design and delivery of a production RAG system (LangChain, ChromaDB) to reduce manual handling of 150+ monthly HR queries; built a labeled evaluation set and iteratively tuned retrieval and prompts.
- Recognizing model improvements were being judged on intuition rather than data, built a structured evaluation framework (precision, recall, F1, confusion matrix) benchmarking every model and prompt iteration against real query distributions, establishing the team's first metrics-driven review process.
- Designed explicit grounding boundaries for a document-scoped QnA feature, constraining the model to answer strictly within user-uploaded course material, eliminating hallucinated, out-of-context responses for student-facing use.

Software Engineer Intern, Ness Digital Engineering, India Nov 2021 – Jun 2022

- Resolved critical transaction latency under high banking load by redesigning Node.js REST APIs and optimizing MySQL + MongoDB schemas, and built CI/CD pipelines (Docker, Jenkins) to replace manual deployments; achieving ∼37% latency reduction, ∼45% faster retrieval, and fewer deployment incidents across sprint cycles.
- Improved frontend performance and reduced bug escape rate by enhancing UI components in React and Angular and authoring unit/integration tests, partnering with QA and senior developers, maintaining zero critical bugs shipped to production.

Education

University of Illinois at Springfield, Master's degree in Computer Science Aug 2022 – Dec 2023
MIT ADT University, Bachelor's degree in Computer Science and Engineering Aug 2019 – May 2022

Publications

Disha Shetiya, Pranita Davankar, Parnesh Adawadkar, Kajal Patil (Jan.–Feb. 2021). Paper: "Smart Refrigerator for Fruit Classification." In: *IOSR Journal of Computer Engineering (IOSR-JCE)*, Volume 23, Issue 1, Ser. I, pp. 07–13. [URL]

Disha Shetiya, Pranita Davankar(Jul.–Aug. 2021). Paper: "Attendance system with mask detection." In: *IOSR Journal of Computer Engineering (IOSR-JCE)*, Volume 23, Issue 4, Ser. I, pp. 16–21. [URL]